

Differentiable transmon design with a tensor-native finite-element electromagnetics solver: gradient-based optimization of charging energy, cross-validated against Palace

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Abstract

Superconducting-qubit electromagnetic design is an iterative guess-and-check loop: the workflow’s production field solvers expose no derivatives of design figures-of-merit, so state-of-practice optimizers iterate time-consuming electromagnetic simulations and inject parameter updates from separate analytic physics models (Eriksson et al., 2025). We demonstrate the missing capability — *solver-derived* design gradients from a 3D finite-element electromagnetics solve — and use it for gradient-based optimization of a transmon’s charging energy E_C . GEODE-FEM is a Rust solver built on the Burn tensor framework; a discrete-adjoint layer bridges the one stage its autodiff tape cannot reach (the sparse factorization), yielding a validated 2×2 sensitivity matrix — material ($\partial/\partial\epsilon$) and geometry ($\partial/\partial X$) derivatives on both the electrostatic and the driven $H(\text{curl})$ operators — each finite-difference-validated with mutation tests (relative errors 10^{-9} – 10^{-5}). Because the capacitance $C = \phi^\top K \phi$ is variationally stationary, its shape derivative collapses to a Hellmann–Feynman-like explicit term; composing it with the capacitance-to- E_C chain, Newton iteration drives a capacitor geometry to a 0.2156 GHz target E_C , confirmed by an independent fresh solve to 1.4×10^{-15} , and on the real 133k-tetrahedron device mesh $\partial C_\Sigma/\partial\theta$ is FD-validated at 1.15×10^{-4} with a genuine two-step convergence to a within-budget target. Harmonic mesh motion extends the safe deformation budget by $8.90\times$, but reaches only 128.793 fF toward the 89.9 fF anchor; the falling sensitivity implies a conditional extrapolated shortfall of at least $6.3\times$ the available budget. Multi-parameter anchor optimization remains pending, and eigenmode/EPR derivatives remain roadmap. The correctness credential: on the identical mesh, GEODE-FEM reproduces all six of Palace’s transmon eigenmodes to $\leq 0.033\%$ (worst case 0.032%). The contribution is a capability, not a speed race — on raw performance the measured record shows no corner where GEODE-FEM beats Palace at scale, and we report it honestly. Every number traces to a committed artifact.

1 Introduction

The hard problem in superconducting-qubit electromagnetic design is the *inverse map*: given target Hamiltonian parameters — a charging energy E_C , an anharmonicity α , coupling capacitances — produce a physical layout. Today that map is walked by hand: parameterize a geometry, run a 3D electromagnetic simulation, extract the parameters, compare, adjust, repeat. The state-of-practice automated workflow makes the loop’s structure explicit: QDesignOptimizer (Eriksson et al., 2025) iterates time-consuming electromagnetic simulations of HFSS-class solvers and guides the parameter updates with *separate*, user-defined analytic physics models — precisely because the field solver at

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the center of the loop produces observables but no derivatives of those observables with respect to the design. The gradient the optimizer actually wants, $\partial(\text{figure of merit})/\partial(\text{geometry, materials})$ through the field solve itself, is structurally unavailable from the production solvers — commercial (HFSS, COMSOL) and open-source (Palace (Palace developers, AWS Center for Quantum Computing, 2023)) alike: their assembly and factorization pipelines are hand-written kernels with no derivative path, and none exposes an adjoint of its solve.

The surrounding literature brackets this gap without closing it. At the *lumped-circuit* level, differentiation is a solved problem: SQcircuit (Rajabzadeh et al., 2023) analyzes arbitrary superconducting circuits, and its gradient-based follow-up (Rajabzadeh et al., 2024) computes eigenvalue and eigenvector gradients of the circuit Hamiltonian via PyTorch autodiff, optimizing qubit designs in circuit-parameter space. But the map from geometry and materials *to* those lumped parameters still runs through non-differentiable 3D electromagnetic simulation. At the *field* level, differentiable electromagnetics is owned by photonics: adjoint and autodiff methods on FDTD and integral-equation solvers are a mature inverse-design industry (Hammond et al., 2022; Hughes et al., 2019; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022), and differentiable FEM at scale has been proven outside electromagnetics by JAX-FEM in solid mechanics (Xue et al., 2023). What none of these supplies is the regime superconducting-qubit (and, more broadly, RF) design lives in: *frequency-domain finite elements on body-fitted tetrahedra with anisotropic materials* — distributed, field-level EM gradients from the same FEM solve that produces the design observables. That wedge is, to our knowledge, uncontested; this paper occupies it.

GEODE-FEM (“geode-fem”) is positioned to close the gap by construction rather than retrofit. It is a full-wave 3D $H(\text{curl})/\text{Nédélec}$ (plus scalar electrostatic) finite-element solver written in Rust on the Burn tensor framework (Tracel AI, 2024): element assembly is batched $[n_{\text{elem}}, 6, 6]$ tensor algebra on an autodiff-capable substrate, so the derivative of assembly with respect to materials and node coordinates is available where a classical solver has only hand-written kernels. One stage breaks the tape: the sparse direct factorization (faer (Kazdadi, 2026)) at the heart of every solve. A discrete-adjoint layer closes exactly that gap — one extra (transpose-)solve, reusing the forward factorization, restores the full gradient (Section 8). The resulting capability is general — shape and material sensitivities for RF, photonic, and electrostatic device design — and the transmon is our demonstration vehicle because its unmet need is the clearest documented in the literature, not because the method is qubit-specific.

A design-gradient claim from a new solver is only as credible as the solver’s forward answers, so the paper carries its correctness credential up front: on a real transmon-plus-readout-resonator geometry meshed once and consumed identically by both codes, geode-fem reproduces Palace — the MFEM-based (Anderson et al., 2021; Andrej et al., 2024) reference solver for superconducting-qubit design — to within 0.033% across all six eigenmodes (worst case 0.032%; junction LC mode 0.001%), with exact analytic tripwires guarding against coincidental agreement (Section 5). This parity is *a credential, not the contribution* — matching Palace is the price of entry — and it is, incidentally, the first independent third-party validation of the Palace/DeviceLayout.jl transmon stack, which has no publication of its own.

Our contributions are:

- **A validated 2×2 discrete-adjoint sensitivity matrix for FEM electromagnetics.** Material ($\partial/\partial\epsilon$) and geometry ($\partial/\partial X$, exact $\partial K/\partial X$ via forward-mode duals through the element kernels) derivatives, on both the scalar electrostatic and the complex driven $H(\text{curl})$ Nédélec operators — each one forward plus one adjoint solve sharing a single factorization, each validated against full-pipeline central finite differences with mutation tests that reject sign, conjugation, and dropped-term errors (Section 8).

- **A differentiable capacitance $\rightarrow E_C$ chain with a variational-stationarity simplification.** Because $C = \phi^\top K \phi$ is stationary in the solved potential, the implicit term vanishes and the shape derivative is a pure explicit-geometry (Hellmann–Feynman-like) term; validated against the analytic parallel-plate law and central finite differences (Section 8.2).
- **Gradient-based optimization demonstrations, honest tiers labeled.** An end-to-end Newton loop that hits a 0.2156 GHz target E_C and is confirmed by an independent fresh solve at 1.4×10^{-15} (the parametrization is affine, so one-step convergence proves the loop, not a hard optimization); a real-device demonstration on the committed 133k-tet mesh with the gradient FD-validated at 1.15×10^{-4} and a genuine two-step convergence; and harmonic mesh motion that widens the safe budget $8.90\times$ but still stalls at 128.793 fF toward the 89.9 fF anchor (Section 9). The multi-parameter anchor experiment is explicitly pending.
- **The correctness credential.** Same-mesh, same-physics, same-junction-model cross-validation against Palace at $\leq 0.033\%$ across all six modes, with oracle-first methodology and honest-physics reporting — the by-construction absent qubit mode and a disclosed, diagnosed spurious mode (Sections 5 and 6) — doubling as the first independent validation of the Palace/DeviceLayout.jl stack.
- **Mode-verified performance context.** Same-box CPU measurements disclose mode selection, field-output and error-estimator costs; they replace the earlier unsupported one-core-versus-eight-rank efficiency claim. CPU-f64 AMS reduces driven-solve time relative to Jacobi, but its iteration and setup costs still grow with refinement (Section 11). Eigenmode/EPR derivatives remain roadmap (Section 10).

Section 2 positions the work; Section 3 presents the substrate and where its tape breaks; Sections 4–6 establish the benchmark problem and the correctness credential; Sections 7–9 are the contribution — the forward LOM chain, the adjoint layer, and the optimization demonstrations; Section 10 is the eigenmode roadmap; Sections 11–13 report performance context, reproducibility, and discussion.

2 Related Work

Superconducting-qubit design optimization. The design loop this paper attacks is documented across the qubit-design software stack. QDesignOptimizer (Eriksson et al., 2025) is the state of practice: physics-guided multi-parameter optimization that wraps non-differentiable EM simulation (HFSS with pyEPR/Qiskit-Metal-class post-processing) and steers it with analytic models, because no gradient is available from the solver. The SQcircuit line (Rajabzadeh et al., 2023, 2024) differentiates at the lumped-circuit level — including eigenvalue/eigenvector gradients of the circuit Hamiltonian via PyTorch — and optimizes in circuit-parameter space; the geometry-to-parameters map remains simulation-bound. The SQuADDS database (Shanto et al., 2024) attacks the same loop by lookup: pre-simulated, experimentally-validated design points. Qiskit Metal’s quasi-lumped methods (Minev et al., 2021b) and the EPR/black-box quantization physics (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) frame the parameter-extraction layer this paper’s LOM chain implements. None of these obtains gradients from the field solve itself; that is the gap Section 8 closes.

Differentiable electromagnetics — owned by photonics. Adjoint/autodiff EM is a mature inverse-design discipline in nanophotonics (Molesky et al., 2018), expressed through structured grids and integral methods: autograd FDFD/FDTD in ceviche (Hughes et al., 2019), foundry-constrained

JAX inverse design (Schubert et al., 2022), the Meep adjoint pipeline (Hammond et al., 2022), the JAX-native FDTD framework (Mahlau et al., 2026), GPU-accelerated autodiff integral-method scattering in TorchGDM (Ponomareva et al., 2025), and differentiable RCWA (Kim et al., 2024). These codes accept rectilinear grids or Green’s-function discretizations; the superconducting-qubit regime — body-fitted tetrahedra, anisotropic sapphire, $H(\text{curl})$ conformity, frequency domain — is exactly where that literature stops. Differentiable FEM has been proven outside EM: JAX-FEM (Xue et al., 2023) (GPU-accelerated, autodiff-native continuum mechanics) and PyTorch-FEA (Liang et al., 2023) in biomechanics; concurrent work on general differentiable Galerkin assembly (TensorGalerkin (Wen et al., 2026), with the torch-sla adjoint sparse-solver library (Chi and Wen, 2026)) independently confirms the substrate-level thesis while reporting nodal magnetostatics rather than full-wave $H(\text{curl})$ EM. To our knowledge, no prior work delivers field-level design gradients from a frequency-domain 3D FEM electromagnetics solve of the class RF/superconducting design uses.

FEM kernels by code generation, and tensor compilers. Generating FEM kernels from high-level descriptions is a two-decade tradition — FFC (Kirby and Logg, 2006), UFL (Alnæs et al., 2014), Firedrake (Rathgeber et al., 2017), TSFC (Homolya et al., 2018) — whose modern GPU expression is libCEED (Brown et al., 2021) and MFEM’s partial-assembly path (Andrej et al., 2024). Palace itself runs on that domain-specific stack. geode-fem’s substrate bet is categorically different: outsource kernels, scheduling, and backend portability to a *general-purpose ML tensor stack* (the Halide/TVM compiler lineage (Chen et al., 2018; Ragan-Kelley et al., 2013), reaching scientific computing via ML frameworks), and phrase assembly and matrix-free application as batched tensor algebra. For this paper the decisive property of that choice is not portability but *differentiability*: an autodiff-capable substrate makes the assembly derivative available by construction, which is what the adjoint layer of Section 8 composes with. Adjacent genera — simulation DSLs (Holl et al., 2020; Hu et al., 2019), neural-ansatz eigensolvers (Jiang et al., 2024), hand-written CUDA FEM (Meng et al., 2014) — trade away either the general-purpose stack or the discretization itself.

Validation and cross-solver benchmarking. Code-to-code comparison on precisely specified problems is an established genre in computational electromagnetics (the TEAM workshop tradition since 1986¹; nano-optics solver comparisons (Hoffmann et al., 2009)); in V&V vocabulary (Oberkampf and Roy, 2010; Roache, 1997) our credential is verification-side, code-to-code evidence. On the qubit-EM axis the closest works are SQDMetal (Sommers et al., 2025) (Palace cross-checked against COMSOL/HFSS at the 0.02–0.13% level, independently constructed models) and the Palace-centered workflow of Ye et al. (2025) (validated against cryogenic measurement at $\sim 0.3\%$). Both wrap and trust Palace; neither is a same-mesh comparison against an independently *implemented* solver. Our protocol eliminates meshing and model entry as confounders — identical MSH fixture, identical lumped-shunt junction, first-order Nédélec on both sides — so residuals isolate formulation and solver correctness. Mature open EM tools normally carry a methods paper (GetDP (Dular et al., 1998), FEniCS/DOLFIN (Logg and Wells, 2010), Gmsh (Geuzaine and Remacle, 2009), scikit-rf (Arsenovic et al., 2022), MFEM (Anderson et al., 2021)); Palace and DeviceLayout.jl are unpublished outliers (AWS Center for Quantum Computing, 2025; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson, 2025), which is the gap the credential narrows from the outside. geode-fem’s numerics: first-order Nédélec elements (Nédélec, 1980), shift-invert Lanczos (Lehoucq et al., 1998) over a real symmetric pencil factorized with faer (Kazdadi, 2026); Palace: SLEPc-class Krylov (Hernandez et al., 2005)

¹The TEAM founding documents are workshop reports without resolvable DOIs; we describe the tradition in prose.

with divergence-free projection. The timed Palace build resolves its default linear solver to SuperLU under GMRES; explicit AMS runs are reported separately (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009); the gauge-fixing literature behind Section 6 is the tree-cotree tradition (Albanese and Rubinacci, 1988; Manges and Cendes, 1995).

3 GEODE-FEM: A Tensor-Native Substrate, and Where the Tape Breaks

geode-fem is written in Rust on the Burn tensor framework (Tracel AI, 2024), whose cubec1 JIT compiles tensor kernels for CPU, CUDA, and WebGPU backends from one source. Every throughput-critical FEM stage is phrased as the batched dense algebra the stack already optimizes.

Batched element-local assembly. For first-order Nédélec tetrahedra the element curl-curl and mass contributions are 6×6 matrices depending on per-element geometry factors and material tensors (including the rotated anisotropic sapphire permittivity of Section 4.1); the scalar electrostatic operator is the analogous P1 kernel. geode-fem computes them for all elements at once as batched $[n_{\text{elem}}, 6, 6]$ tensor operations — one kernel launch per assembly stage rather than an element loop — then either scatters into a global sparse matrix (the assembled f64 CPU path) or retains the batch for matrix-free application.

Matrix-free application and on-device Krylov. The matrix-free operator apply is a three-stage tensor pipeline — gather global DOFs to element-local vectors, batched matmul against the stored element operators, scatter-add back — and the driven frequency-sweep solve runs a conjugate orthogonal conjugate gradient (COCG) iteration entirely on the device with a constant number of scalar host-device synchronizations per iteration. The backend (ndarray on CPU, CUDA on GPU) is a type parameter, not a code path. Section 11 reports honestly how far below the GPU-amortization threshold the current benchmark sizes sit.

Honest constraints. Two boundaries are constraints, not footnotes. The CUDA backend (burn-cuda 0.21) is f32-only today — the underlying cubec1 runtime currently disables f64² — so every GPU result is mixed-precision-qualified. And the eigensolve that produces the credential comparison is factorization-bound: its inner solves are faer sparse-LU factorizations (Kazdadi, 2026), a classical serial CPU component outside the tensor-compiler story.

Where the tape breaks — and why an adjoint layer, not autodiff-through-the-solver.

The substrate choice has a consequence the rest of this paper builds on: Burn’s automatic differentiation reaches everything *up to* the linear solve — material and node-coordinate dependence of the assembled operators is differentiable by construction — but the sparse factorization is an external, opaque kernel that breaks the tape. Differentiating *through* a factorization by unrolling is neither practical nor necessary: the classical discrete-adjoint identity turns the missing derivative into one additional linear solve against the *same* factorization (Section 8). The division of labor is deliberate — autodiff where the substrate provides it (element kernels, via forward-mode duals), adjoint algebra where it does not (the solve) — and it is what makes the gradients in this paper *solver-derived* in the sense the design loop of Section 1 lacks: the same assembly and the same factorization that produce the observable produce its gradient.

²Upstream constraint of the burn/cubec1 stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

4 Benchmark Problem: Geometry, Mesh, and Junction Model

4.1 Geometry and mesh provenance

The device is the `SingleTransmon` example shipped with `DeviceLayout.jl` v1.15.0 ([AWS Center for Quantum Computing, 2025](#)): a transmon with readout resonator on a sapphire substrate, enclosed in a vacuum box. The sapphire is anisotropic, with permittivity tensor $\varepsilon = R \text{diag}(9.3, 9.3, 11.5) R^\top$ rotated approximately 36.87° in-plane. The `DeviceLayout.jl` workflow emits seven named physical groups (metal, substrate, vacuum, junction lumped-element surface, two readout ports, exterior boundary) and drives Gmsh ([Geuzaine and Remacle, 2009](#)) to produce a tetrahedral mesh, exported as an MSH 4.1 fixture and pinned by SHA-256 (Section 12).

The mesh has 22,684 nodes and 133,314 tetrahedra, yielding 156,863 Nédélec edge unknowns, of which 133,108 interior degrees of freedom remain after eliminating perfect-electric-conductor (PEC) boundary edges. PEC is applied on all metal surfaces and the exterior box; the two readout ports are left open (lossless), so no resistive element enters and the eigenproblem pencil stays real symmetric. Figure 1 shows the geometry with physical groups color-coded. Every measurement in this paper — the eigenmode credential, the electrostatic LOM chain, and the real-device gradient demonstration — runs on this one committed mesh (plus a uniformly refined variant for the scale study of Section 11).

4.2 The junction model: a lumped reactive shunt

The Josephson junction is modeled identically in both solvers as a lumped reactive shunt — a parallel $L \parallel C$ element with $L = 14.860 \text{ nH}$ and $C = 5.5 \text{ fF}$ — attached to the four-triangle `lumped_element` surface group Γ . Following the uniform lumped-port construction, with port length ℓ and width w , the shunt contributes

$$K_{\text{port}} = \frac{\ell}{w L} S_\Gamma \quad \text{and} \quad M_{\text{port}} = \frac{C \ell}{w} S_\Gamma, \quad (1)$$

where S_Γ is the port-surface matrix assembled on Γ . The eigenproblem is then the real symmetric generalized pencil

$$(K + K_{\text{port}}) x = \omega^2 (M + M_{\text{port}}) x. \quad (2)$$

In Palace, the same physics is entered as a reactive `LumpedPort` ($L \parallel C$, no resistance) on the same surface group; the readout ports are configured open, so Palace’s pencil is likewise real. This one-to-one correspondence is what makes the comparison same-physics as well as same-mesh.

5 The Correctness Credential: Same-Mesh Cross-Validation Against Palace

A solver claiming design gradients must first prove its forward answers; this section is that proof. Palace’s own agreement with commercial tools is established ([Sommers et al., 2025](#)), so reproducing Palace is the bar a new implementation must clear before anything it computes *about* its solutions (such as their derivatives) is worth discussing.

Protocol. The benchmark is oracle-first. The mesh is generated once, committed, pinned by SHA-256, and consumed by both solvers at matched (first) order with the identical junction model; residual disagreement is therefore attributable to formulation or implementation, not meshing. Two exact analytic tripwires guard against coincidental or self-consistent-but-wrong agreement: the isolated junction LC resonance $f_{LC} = 1/(2\pi\sqrt{LC}) = 17.60 \text{ GHz}$ anchors the junction mode, and a Josephson $\sqrt{L}$ -scaling test requires that doubling L move the junction mode by exactly $1/\sqrt{2}$ — measured: $17.4901 \rightarrow 12.37 \text{ GHz}$, committed ratio 0.7071 (computed from the unrounded values; Figure 3). The suite also contains inverse tests that *must fail* (perturbed configurations required

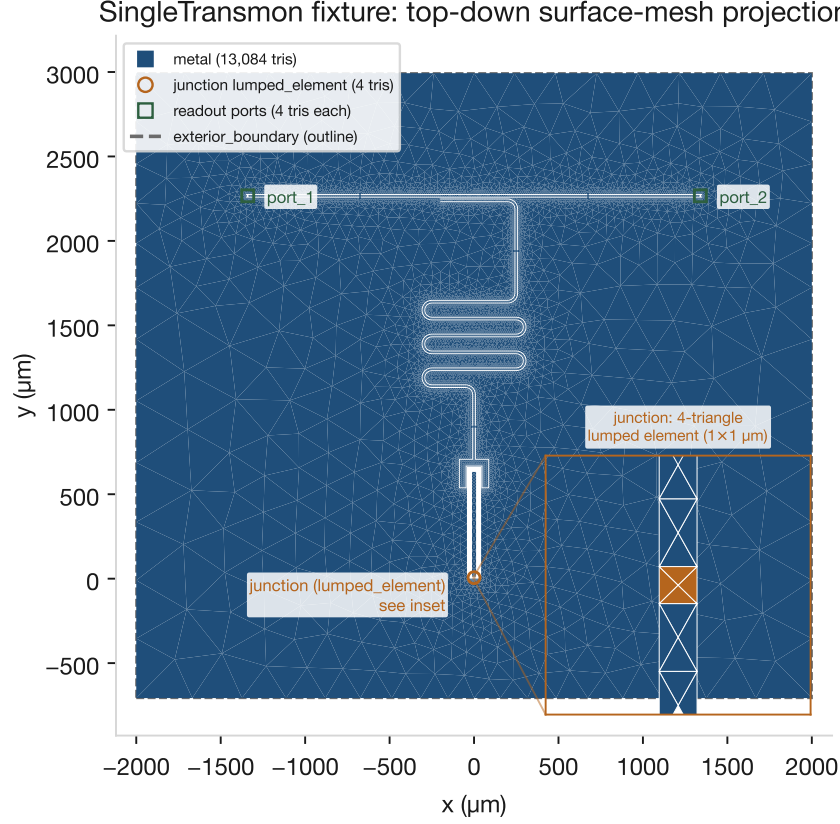


Figure 1: Benchmark geometry: top-down projection of the committed surface mesh (the SHA-256-pinned MSH 4.1 fixture) for the DeviceLayout.jl v1.15.0 `SingleTransmon` transmon and readout resonator. Surface physical groups are color-coded — the metal ground plane (with the coplanar-waveguide feedline, meander resonator, and transmon slot as etched gaps), the four-triangle junction `lumped_element` surface (enlarged in the inset, actual triangulation shown), the two readout ports, and the exterior-boundary outline; the `substrate` and `vacuum` volume groups fill the enclosing box. Both solvers consume this identical mesh (22,684 nodes; 133,314 tetrahedra).

to disagree with the oracle — a pipeline that still passes under them is not measuring anything), and the Palace oracle is deterministic: a rerun from the committed configuration reproduced its committed `eig.csv` bit-for-bit.

Mode identification. Modes are identified, not eyeballed. Within geode-fem, the stiffness-participation ratio

$$p = \frac{x^\top K_{\text{port}} x}{x^\top (K + K_{\text{port}}) x} \quad (3)$$

cleanly separates the junction LC mode ($p = 1.000$) from resonator and box modes ($p = 0.000$ at the artifact’s three-decimal precision). Palace reports a field-based per-port energy participation, a differently normalized quantity that does *not* rank the modes the same way on the committed run (all six of its port-EPR values are $\leq 5 \times 10^{-4}$, with the junction mode’s the smallest in magnitude); the two diagnostics are complementary, not a shared discriminant, and we do not compare them numerically. Cross-solver mode identification therefore rests on frequency agreement against the

Table 1: Eigenmode agreement on the identical mesh (22,684 nodes / 133,314 tets / 156,863 Nédélec DOFs, 133,108 interior after PEC), first order in both solvers, junction as a lumped reactive shunt ($L = 14.860$ nH, $C = 5.5$ fF). geode-fem at commit 3174015; Palace at commit fba6a5b. All values are quoted verbatim from the committed `benchmarks/transmon_eigen/results.toml`: geode-fem frequencies at the committed three decimals, Palace at full precision, and the committed per-mode `rel_err_pct`, computed against the rounded geode-fem values (precision disclosure in Section 12).

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
<code>resonator</code>	readout resonator	5.153	5.151335830348	0.032
<code>mode_2</code>	box/cavity	15.465	15.46052107794	0.029
<code>junction_lc</code>	junction LC	17.490	17.49010903536	0.001
<code>mode_4</code>	box/cavity	18.693	18.69165792915	0.007
<code>mode_5</code>	box/cavity	20.703	20.69755679425	0.026
<code>mode_6</code>	box/cavity	26.088	26.08089940472	0.027

analytic anchor.

Result. Table 1: all six modes agree to $\leq 0.033\%$ (worst case 0.032%), with the junction LC mode — the mode the junction model exists to produce — at 0.001%. The junction mode sits 0.6% below the isolated-circuit anchor $f_{LC} = 17.60$ GHz in *both* solvers (the small shift reflects its embedding in the field problem), and the agreement is between genuinely independent code paths — a batched-tensor assembly feeding a direct-factorization Lanczos in Rust, and a projected, preconditioned Krylov method in C++ — so a common implementation error is an unlikely explanation for the residual, and the tripwires exclude self-consistent-but-wrong junction physics.

6 Honest Physics: The Absent Qubit Mode and a Disclosed Spurious Mode

6.1 No ~ 4 GHz qubit mode exists, by construction

The DeviceLayout.jl blog workflow (Peairs and Carson, 2025) reports a spectrum running from 4.14 to 5.591 GHz — including a ~ 4 GHz qubit mode — while Table 1 contains no mode below 5.15 GHz. This is a property of the model, stated plainly: the physical transmon frequency arises from the Josephson inductance resonating against the *pad (shunt) capacitance* of roughly 80–100 fF (Koch et al., 2007), and the lumped-shunt model of Section 4.2 gives the junction only its own 5.5 fF, so its resonance lands at 17.5 GHz. Both solvers agree on the absence: Palace’s projected eigensolve, instructed to hunt at $\sigma = 4.5$ GHz, finds nothing below the 5.15 GHz resonator. The blog spectrum is not reproduced by this model, and no agreement with it is claimed. The route to qubit quantities from this geometry is the electrostatic LOM chain of Section 7 — which is exactly the chain this paper makes differentiable.

6.2 One spurious mode: disclosed, diagnosed, characterized

Honest cross-validation requires reporting where the solvers *disagree*, and there is exactly one such place. geode-fem’s headline Lanczos operates on the raw pencil of Eq. 2 without a gauge projection, and on this problem it leaks one spurious mode at 3.4528 GHz, junction-localized ($p = 0.9942$), with no Palace counterpart. The artifact was run to ground through a measured three-step gauge/projection arc, every step committed (Section 12): (1) classical tree-cotree elimination (Albanese and Rubinacci, 1988; Manges and Cendes, 1995) removes it but is not spectrum-preserving for the generalized eigenproblem — the resonator shifts 1.64% ($5.1528 \rightarrow 5.2372$ GHz), outside our bar, pinned as a regression test; (2) the spectrum-preserving bulk divergence-free projector

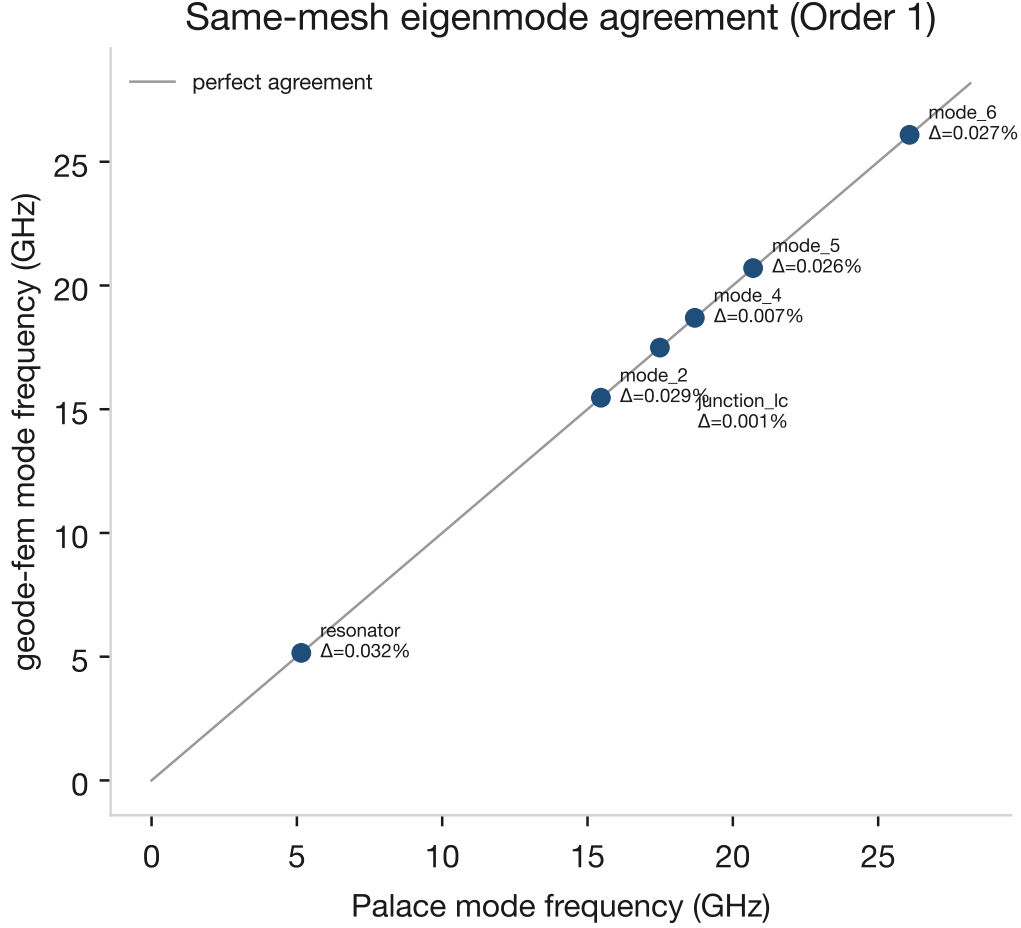


Figure 2: Mode-frequency agreement: geode-fem versus Palace on the identical mesh, with per-mode relative deviations annotated. Data from the committed benchmark artifacts (`benchmarks/transmon_eigen/results.toml` and Palace’s committed `eig.csv`).

(structurally Palace’s confinement) preserves the cavity spectrum (0.029%) and collapses the 13,747-mode gradient cluster, but *deflates the physical junction mode* — a lumped inductor is a quasi-static, curl-free flux path (measured divergence ratio 50.2; projected norm 1.06×10^{-4}) — while the spurious mode, being genuinely solenoidal (divergence ratio $\approx 6 \times 10^{-15}$), *survives*; (3) a port-aware rank-1 re-admission of the junction flux direction yields a projector that retains all six physical modes at $\leq 0.029\%$ worst case versus Palace. What the artifact *is*: under the L-doubling harness it moves $3.4528 \rightarrow 2.4449$ GHz (ratio 0.7081, the junction $1/\sqrt{L}$ law) with 99.4% of its stiffness energy in the K_{port} surface term — a second LC-like resonance of the port-surface operator S_{Γ} , which Palace’s **LumpedPort** elimination represents differently and therefore never sees. The headline comparison deliberately remains the committed ungauged benchmark (the port-aware solver shipped after the headline measurement); the filter now excludes a *characterized* artifact rather than an unexplained one. Figure 4 contrasts the spectra. The arc measures precisely what Palace’s divergence-free projection buys in a production eigensolver, and what a lumped-element formulation demands beyond it.

The subsequent port-subspace implementation removes the near-zero survivor and admits the junction direction with one shifted factorization. On this fixture, requesting seven modes at 4.5 GHz

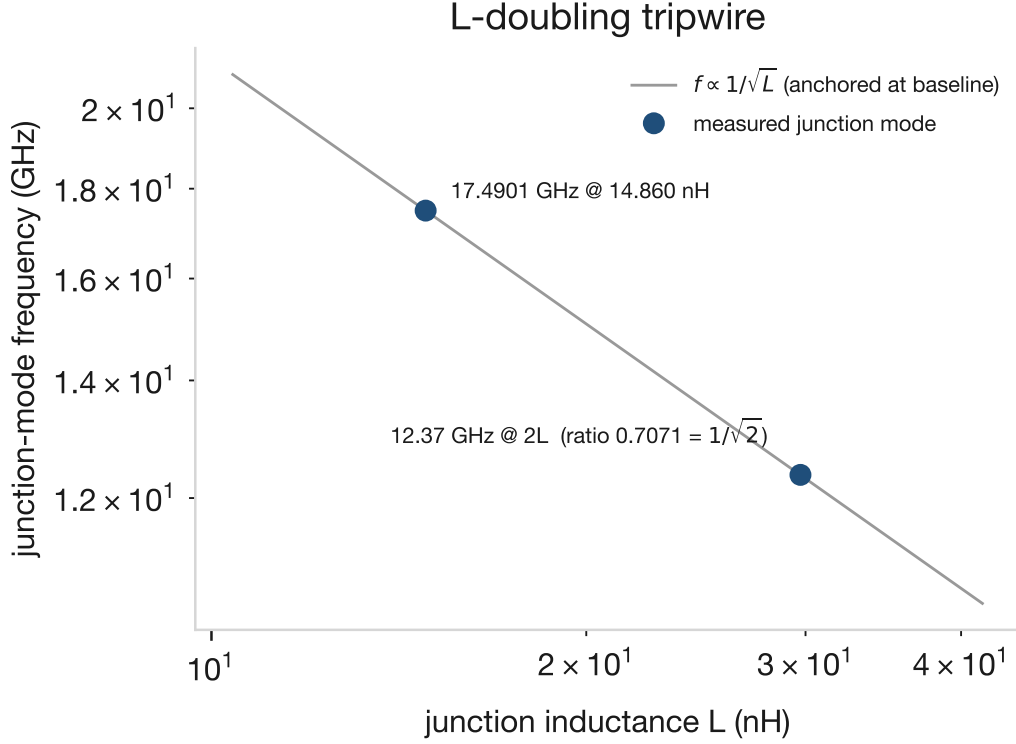


Figure 3: Josephson inductance-scaling tripwire: junction-mode frequency versus junction inductance L on log-log axes. Doubling L moves the junction mode from 17.4901 to 12.37 GHz — committed ratio $0.7071 = 1/\sqrt{2}$ — exactly as $f \propto 1/\sqrt{L}$ requires. Measured values from the committed `benchmarks/transmon_eigen/results.toml` tripwire block.

returns the six physical modes (worst relative frequency error 0.0289%) plus the 3.452751 GHz port mode; requesting six still misses the highest physical mode. The committed local comparison records mean process CPU times of 46.32s for the older eight-mode extraction route and 23.25s for the new seven-mode route. These are geode-only measurements on a loaded Apple M3 Ultra, not a Palace comparison or a same-box update to the EC2 timings (Section 11.1). The extra mode remains a genuine eigenpair of the distributed reactive surface shunt. Removing a gradient kernel cannot remove it, and high port participation cannot distinguish it from the physical junction mode. A revised port formulation or a principled reference-free classifier remains unresolved; frequency matching to Palace is still used to identify the physical modes.

7 The Forward LOM Chain: From Fields to E_C

The lumped-oscillator-model (LOM) branch computes the transmon’s Hamiltonian parameters from the *electrostatic* problem on the same committed mesh — and it is the chain the remainder of this paper differentiates. The mesh’s metal is split into three node-disjoint conductors (ground+resonator, feedline, island); a tensor- ϵ electrostatic solve yields the Maxwell capacitance matrix by the energy-reaction form $C_{ij} = \phi_i^T K \phi_j$; the island self-capacitance is reduced for the floating feedline ($C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$); and the charging energy follows as $E_C/h = e^2/(2C_\Sigma h)$, with the transmon spectrum evaluated charge-basis-exactly after Koch et al. (2007).

Measured on the committed fixture (`benchmarks/transmon_quantum/results.toml`): $C_\Sigma = 136.7$ fF, hence $E_C = 0.142$ GHz, $E_J/E_C = 77.6$ (with E_J from the junction inductance), $\omega_{01} =$

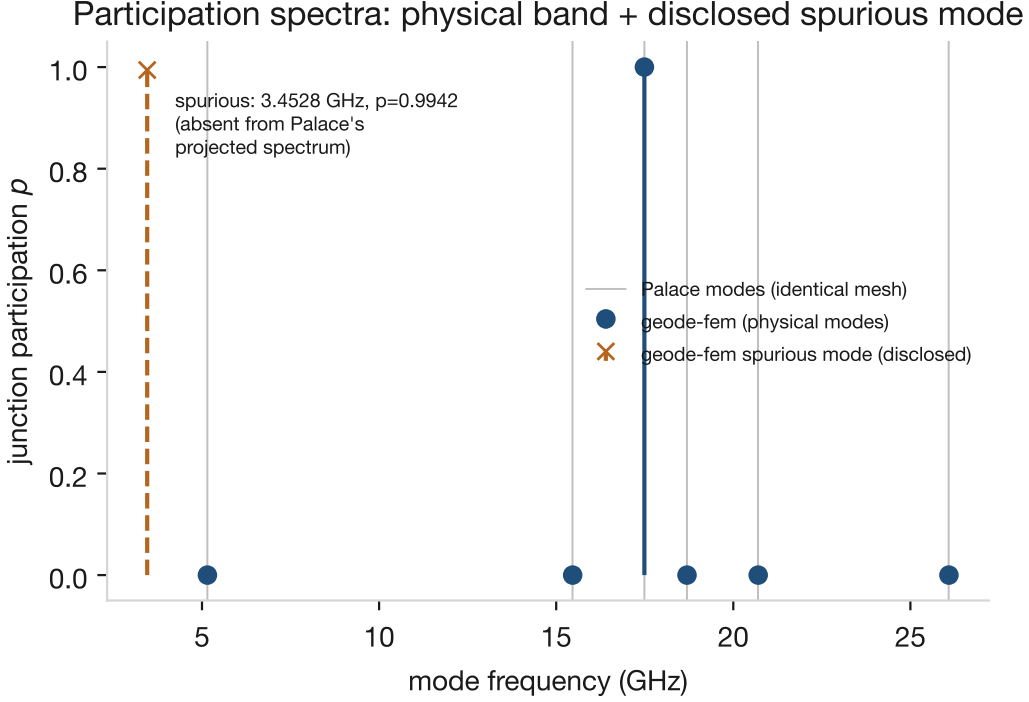


Figure 4: geode-fem junction-participation spectrum on the identical mesh, with Palace’s mode locations (from the committed `eig.csv`) overlaid. The physical junction LC mode appears at 17.49 GHz in both solvers’ spectra (geode-fem stiffness participation $p = 1.000$); geode-fem’s un-projected headline Lanczos additionally leaks one spurious junction-localized mode at 3.4528 GHz ($p = 0.9942$) with no Palace counterpart — characterized in Section 6.2 as a port-surface artifact obeying the junction $1/\sqrt{L}$ law. Palace’s field-based port-EPR is a differently normalized diagnostic and is not plotted on geode-fem’s participation axis (Section 5).

3.38 GHz, and anharmonicity $\alpha = -0.158$ GHz — a plausible transmon computed end-to-end from committed geometry, with the $\alpha \approx -E_C$ sanity holding.

The honest negative that travels with this chain. The extracted $C_\Sigma = 136.7$ fF *exceeds* the ~ 90 fF design-anchor value back-solved from the blog workflow’s expected $E_C = 0.2156$ GHz. The committed benchmark diagnoses this as a model-definition difference, not a truncation artifact (grounding the exterior wall moves C_Σ by a relative 6×10^{-5} ; the scalar-versus-tensor sapphire treatment shifts it 0.75%). We report the extracted numbers and the discrepancy; we do not retrofit to the anchor. Section 9 returns to this gap as an optimization target — and reports honestly how far a fixed-topology parametrization gets toward closing it.

8 Differentiable Sensitivities: The Discrete-Adjoint Layer

8.1 The 2×2 sensitivity matrix

For a solve $A(\varepsilon, X)x = b(X)$ and a scalar observable $g(x)$, the discrete adjoint $A^\top \lambda = \partial g / \partial x$ turns the parameter gradient into explicit contractions:

$$\frac{dg}{d\varepsilon_k} = -\lambda^\top \frac{\partial A}{\partial \varepsilon_k} x, \quad \frac{dg}{dX} = \lambda^\top \frac{\partial b}{\partial X} - \lambda^\top \frac{\partial A}{\partial X} x \quad (4)$$

Table 2: The merged 2×2 discrete-adjoint sensitivity matrix. Each cell: one forward + one adjoint solve sharing a single factorization; element Jacobians exact (forward-mode duals through the production kernels); validated against central finite differences of the full pipeline, with mutation tests. “FD rel-err” is the observed agreement of the analytic gradient with the FD reference in the merged test suite (test acceptance bounds are looser; see the module sources). Modules in `crates/geode-core/src/`.

Operator	Parameter	Module (issue / PR)	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\epsilon$	<code>adjoint</code> (#570 / #573)	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	<code>shape</code> (#571 / #575)	$\sim 1 \times 10^{-9}$
driven $H(\text{curl})$	material $\partial/\partial\epsilon$	<code>driven::adjoint</code> (#576 / #579)	2.3×10^{-5} (worst region)
driven $H(\text{curl})$	geometry $\partial/\partial X$	<code>driven::shape</code> (#577 / #581)	2.2×10^{-9} , 1.2×10^{-8} (two maps)

(real case; the complex driven case takes $2 \operatorname{Re}[\cdot]$ of the same contractions with the *transpose*, not conjugate-transpose, solve — the pencil is complex-symmetric, so the adjoint solve reuses the forward LU factorization). Everything on the right-hand side is assembly-level and therefore differentiable on the tensor substrate: the element Jacobians $\partial A/\partial X$ and $\partial b/\partial X$ are *exact*, computed by forward-mode dual numbers through the same closed-form element kernels the solver assembles — no finite-difference truncation enters the gradient. Each gradient costs one forward plus one adjoint solve sharing a single factorization (asserted programmatically: `n_factorizations = 1`).

Four adjoint modules are merged and validated, spanning material and geometry sensitivities on both operators (Table 2). Each carries a committed test that the analytic gradient match a central finite difference *of the full independent pipeline* (perturb the parameter, re-assemble, re-solve through the public solver path, recompute g) — and, load-bearingly, mutation tests proving the tolerances bite: the classic complex-adjoint conjugation error (A^H for A^T) is rejected by the FD check, as is a dropped term. The driven-shape case surfaced a physics subtlety worth recording: the current-source right-hand side $b = i\omega\mu_0 \int N \cdot J$ is geometry-dependent, so the shape gradient carries a $\partial b/\partial X$ term absent from the material case — omitting it fails the finite-difference cross-check outright.

The matrix is deliberately 2×2 : material and geometry are the two design-parameter classes, and the scalar and driven operators are the electrostatic (LOM) and RF (S-parameter/EPR-workhorse) branches of the solver. The LOM optimization of Section 9 exercises the scalar column end-to-end; the driven column is merged and validated at the operator level and composes with the driven solve that geode-fem’s GPU path accelerates — RF shape optimization is the natural follow-on, not exercised here.

8.2 The capacitance $\rightarrow E_C$ chain, and a vanishing adjoint

Composing the shape adjoint with the LOM chain of Section 7 gives $\partial(C, E_C, \alpha)/\partial(\text{geometry})$ (merged as `shape::capacitance_shape_gradient`, #583 / PR #586). The composition carries an elegant simplification. The capacitance observable is the stored field energy, $C = 2W = \phi^T K(X)\phi$, and W is *variationally stationary* in the solved potential ϕ : at the forward solution the free rows of the cotangent $K\phi$ vanish (equilibrium), so the adjoint solution $\lambda \approx 0$ and the entire shape derivative collapses to the explicit-geometry term

$$\frac{dC}{dX} = \phi^T \frac{\partial K}{\partial X} \phi \quad (\text{at the solution}), \quad (5)$$

a Hellmann–Feynman-like result: no adjoint back-solve is needed for the capacitance itself. The implementation keeps the general adjoint form (the mutation test measures the implicit part at $< 10^{-8}$ of the total, confirming the collapse numerically rather than assuming it), and chains

through $\partial E_C/\partial C_\Sigma = -e^2/(2C_\Sigma^2 h)$ exactly. Validation is double: against the analytic parallel-plate law ($\partial C/\partial d = -\varepsilon_0\varepsilon_r$ per unit area under a gap scaling; $\partial C/\partial A = +2\varepsilon_0\varepsilon_r$ under an area scaling — opposite signs, both matched) and against central finite differences of the full pipeline, with observed relative errors at the 10^{-9} level.

9 Gradient-Based Optimization: Demonstrations and an Honest Negative

Two demonstrations, in deliberately labeled honesty tiers: a closed-loop proof on an analytic fixture, then the real device mesh. Both use damped Newton on $E_C(\theta) - E_C^{\text{target}}$ (equivalently $C(\theta)$) with the analytic $dE_C/d\theta$ from Section 8.2; each Newton step costs one forward plus one adjoint solve on a single factorization. The framing claim is capability and step-count, not wall clock: a derivative-free loop (the parameter sweeps of current practice) spends N_{params} extra forward solves per step to estimate what the adjoint delivers exactly for one. No timing comparison against HFSS/Palace-based optimizers is measured or claimed.

9.1 Loop proof: Newton to a target E_C on a parallel-plate fixture

On a parallel-plate capacitor fixture ($\varepsilon_r = 3$; 343 nodes, 1,296 tets; θ scales the gap, so $d = L(1 + \theta)$), Newton iteration drives the design from $C_0 = 120.0$ fF ($E_C = 0.1614$ GHz) to the design-anchor target $E_C = 0.2156$ GHz. Full Newton converges in *one* step — and we state the reason rather than dress it up: $C \propto 1/(1 + \theta)$ makes $E_C(\theta)$ *affine* in θ , so one-step convergence is expected, and this tier proves the loop end-to-end (gradient, chain rule, update, convergence test), not a hard optimization. The converged $\theta = 0.335657909$ matches the closed-form optimum to 3×10^{-15} . The honesty check that matters: a *fresh, independent* forward solve at the converged geometry — assemble, solve, extract, no adjoint machinery — reproduces the target E_C to a relative error of 1.4×10^{-15} . A damped run ($\alpha = 0.5$) converges in 13 steps and supplies the descent illustration (Figure 5a). At the hit target, the Koch-exact spectrum gives $E_J/E_C = 51.0$, $\omega_{01} = 4.128$ GHz, $\alpha = -0.247$ GHz — the ~ 4 GHz qubit regime the design anchor implies. All trajectory values are committed in `benchmarks/transmon_diffopt/results.toml`.

9.2 The real device: validated gradients and deformation limits

The second tier runs on the committed 133k-tet SingleTransmon mesh itself, with a fixed-topology shape parameter θ that scales the island conductor in-plane about its centroid (101 island nodes move; the mesh connectivity is fixed). Three results, in descending order of comfort (`benchmarks/transmon_diffopt/pad_results.toml`):

The gradient is real on the real mesh. At $\theta = 0$ the adjoint gradient $\partial C_\Sigma/\partial\theta = 198.198$ fF is validated against a central finite difference of the *full independent pipeline* (move the island nodes, re-assemble, re-solve, re-extract): relative error 1.15×10^{-4} at $h = 10^{-4}$, with a clean $O(h^2)$ truncation sweep ($1.2 \times 10^{-2} \rightarrow 1.0 \times 10^{-3} \rightarrow 1.15 \times 10^{-4} \rightarrow 1.0 \times 10^{-5}$ as h falls $10^{-3} \rightarrow 3 \times 10^{-5}$). This is the paper’s load-bearing real-geometry validation: the same machinery that passed at 10^{-9} -level on analytic fixtures holds up on production geometry.

The loop genuinely converges on the real mesh. A demonstration target within the mesh’s validity budget ($C_\Sigma = 137.0$ fF) converges in two Newton steps — genuinely non-affine now (the gradient changes by $\sim 21\%$ along the path) — and the converged design is confirmed by a fresh, independent multi-conductor extraction at relative error 5.6×10^{-6} (Figure 5b). The entire real-device run (validation sweep plus both optimization runs) takes 10.7 s.

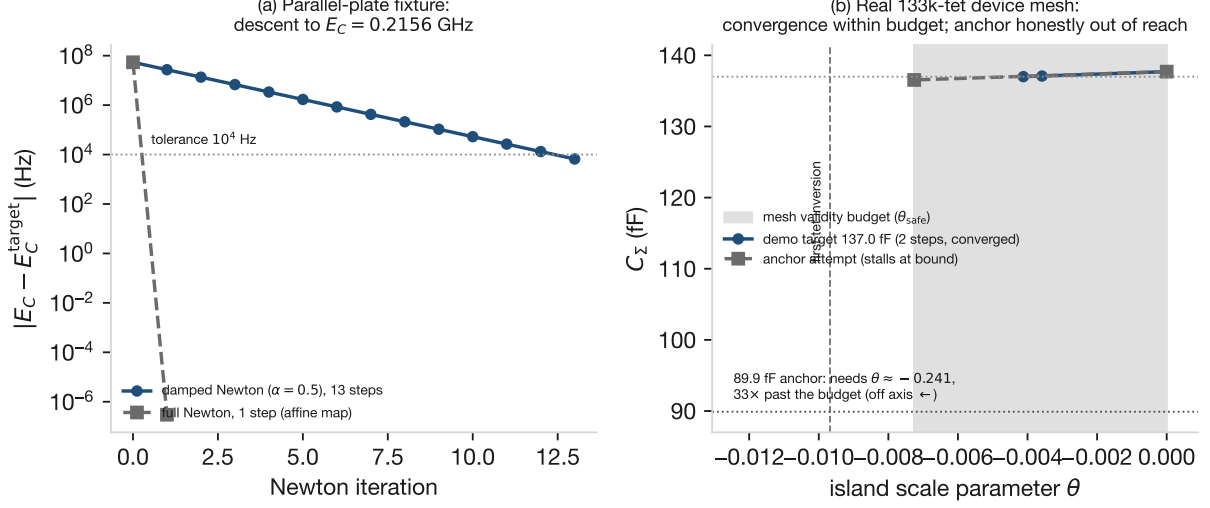


Figure 5: Gradient-based optimization to a target E_C via the discrete-adjoint shape gradient. (a) Parallel-plate loop proof: $|E_C - E_C^{\text{target}}|$ versus Newton iteration; full Newton converges in one step (the parametrization is affine — expected, and disclosed), the damped $\alpha = 0.5$ run supplies the 13-step descent curve; the converged design is confirmed by an independent fresh forward solve at relative error 1.4×10^{-15} (`benchmarks/transmon.diffopt/results.toml`). (b) The real 133k-tet DeviceLayout mesh under the fixed-topology island-scale parameter θ : a within-budget demonstration target ($C_\Sigma = 137.0$ fF) converges in two genuine (non-affine) Newton steps, fresh-solve confirmed at 5.6×10^{-6} ; the bounded run toward the 89.9 fF design anchor stalls honestly at the mesh validity budget $\theta_{\text{safe}} = -0.0073$ (first tetrahedron inversion at $\theta = -0.0097$), $33\times$ short of the $\theta \approx -0.241$ the anchor requires (`benchmarks/transmon.diffopt/pad.results.toml`).

The honest negative: the anchor is out of this parametrization’s reach. The natural target is the ~ 90 fF anchor of Section 7 (89.9 fF; $E_C = 0.2156$ GHz — the last digit differs from the 0.2156 GHz quoted elsewhere because this run targets the rounded 89.9 fF rather than the back-solved 89.843 fF; both values are artifact-faithful). The $\theta = 0$ gradient puts it at $\theta \approx -0.241$ — but the fixed-topology map inverts its first tetrahedron at $\theta = -0.0097$, and the run’s conservative budget (minimum tet-volume ratio 0.25) caps $\theta_{\text{safe}} = -0.0073$, where the bounded Newton run stalls at $C_\Sigma = 136.5$ fF: a $33\times$ shortfall. The diagnosis is geometric, not a gradient failure: the island’s junction-attachment nodes sit $\sim 225 \mu\text{m}$ from the island centroid, so the centroid scale drags them against $\sim 0.7 \mu\text{m}$ junction-region tetrahedra. The $33\times$ figure is a local linear estimate for the rigid island-only map, not a measured path to the anchor.

Harmonic motion extends the budget, but does not reach the anchor. Extending the prescribed island velocity into the volume with a P1 Laplace solve spreads the deformation; the other conductors and far boundary have zero velocity. The field D is computed once, giving $X(\theta) = X_0 + \theta D$. At the same minimum volume-ratio threshold 0.25, the safe negative step grows from -0.007258 to -0.064609 , an $8.90\times$ extension. The harmonic field takes 0.34 s to solve; the validation and bounded optimization run takes 7.8 s. The initial derivative is 209.087 fF per unit θ , with a full-pipeline central-FD relative error 1.054×10^{-4} at $h = 10^{-4}$. Bounded Newton reaches 128.792998 fF, independently confirmed by a fresh extraction, and stalls at the safe boundary, leaving 38.892998 fF to the anchor.

The initial-slope extrapolation in the artifact gives $\theta_{\text{anchor}} = -0.2286$, or $3.54\times$ the harmonic budget. That estimate understates the shortfall because the capacitance sensitivity falls along the

measured path. From $dC/d\theta = -(C/E_C) dE_C/d\theta$, the two recorded steps give 209.1 and 114.5 fF per unit θ . *If the sensitivity does not recover beyond the measured endpoint*, the remaining gap needs at least another $38.893/114.5 \simeq 0.340$ in $|\theta|$, or a total approximately 0.404: at least $6.3\times$ the safe budget after rounding. This is a conditional extrapolation, not a measured bound on an uncomputed continuation. Both runs use scalar trace-averaged sapphire; neither establishes the tensor-material shape chain.

9.3 Multi-parameter real-device anchor experiment (pending)

[**PENDING issue-1034**] The anchor-reaching experiment will combine junction-pinned length, width and cutout-gap parameters, then re-solve the mesh-motion field at each bounded optimization step. A subsequent coupling constraint will test the island-claw ratio $\beta = C_g/C_\Sigma$. The trajectory, parameter bounds, finite-difference checks, minimum element-volume ratios and independent final extraction are not yet available in this draft. No anchor-reaching or two-target result is claimed here.

10 Roadmap: Differentiating the Eigenmode Branch

The LOM branch above differentiates an *electrostatic* solve. The transmon’s dynamical figures-of-merit — resonator frequency, participation-ratio and EPR self/cross-Kerr (Minev et al., 2021a; Nigg et al., 2012) — live in the *eigenmode* problem, and that branch is **not differentiable yet** in geode-fem. The paper commits to the explicit split:

Branch	Observables	Differentiable?
LOM (electrostatic)	E_C , $\alpha \approx -E_C$, coupling C	yes (Sections 8–9)
eigenmode / EPR	resonator f , participation, Kerr	not yet (this section)

Differentiating an eigenpair needs the adjoint-eigenpair (Hellmann–Feynman / Nelson) formulas (Nelson, 1976), which require solving the interior eigenproblem robustly at the physical shift — and geode-fem’s matrix-free interior eigensolve is blocked at exactly that shift. The blocker is measured and committed (`benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`): at $\sigma = 4.5$ GHz on the 133k-DOF fixture, a *single* inner AMS-preconditioned MINRES solve (outer Lanczos step 0) never converges — it plateaus on a nearly flat residual tail below $\sim 10^{-5}$ and was still running at 14,300 inner iterations ($\|r\| \approx 2.6 \times 10^{-6}$) when killed. The plateau is *coarse-solve-invariant* — an exact coarse solve stalls the same way — so the limiter is the SPD-proxy preconditioner $K + |\sigma|M$ itself, not the coarse solve. Nor is there a single inner tolerance that is both cheap and correct: at an inner tolerance of 10^{-2} the entire eigensolve completes in ~ 56 s but returns the non-physical $\lambda \approx 0$ gradient near-kernel cluster (0.42–0.64 GHz), not the physical band. The named path — not built, cited as method, not result — is a preconditioned Helmholtz-projected Jacobi–Davidson iteration, applying the gradient-kernel projection inside the correction equation, for which Liang et al. (2026) provide the current adaptive-multilevel treatment of exactly this singular Maxwell eigenproblem; combined with the Nelson/Hellmann–Feynman eigenpair derivatives, that is the route to differentiable eigenmode/EPR quantities. Until it lands, no claim in this paper extends to derivatives of the eigenmode spectrum.

11 Performance Context

The timings below qualify the cost of the capability. Eigenmode, driven CPU and driven GPU results are separate experiments; none measures the optimizer’s cost or establishes a general solver ranking.

Table 3: Mode-verified CPU timings on one m6i.4xlarge: median wall seconds / measured core-seconds. All rows contain the six physical modes. Three repeats per cell except Palace one-rank Save=0 (one run). The two oracle-tuned requests are marked †; the historical port-aware route is the pre-port-subspace implementation. Palace always requests six modes above 4.5 GHz.

Request	Extra modes	1 wide	8 wide
geode 30 at 4.5 GHz [†]	24	34.45 / 34.44	29.08 / 40.71
geode 6 at 20 GHz [†]	0	33.21 / 33.20	27.26 / 38.87
geode port-aware 8 at 4.5 GHz	2	67.26 / 67.25	51.12 / 91.72
Palace, Save=6	0	186.84 / 186.82	33.35 / 266.66
Palace, Save=0	0	113.00 / 112.98	23.64 / 188.89

11.1 CPU eigenmodes: same box, verified mode content

The earlier timing request, six modes at 4.5 GHz, did not establish equivalent work. In the mode-logged re-run it returns four near-kernel modes, the spurious port mode and only one physical mode in geode-fem, whereas Palace returns all six physical modes. The historical run has no committed mode log or CPU-time record; its mode content and actual thread count cannot be recovered. We therefore withdraw the earlier claim that one geode core beats eight Palace ranks with roughly twelve times fewer core-seconds. This does not change the separate frequency agreement credential of Section 5.

The replacement experiment ran on one otherwise idle EC2 m6i.4xlarge in us-east-1 on 2026-10-08, with geode 61e8571e and Palace fba6a5b. One-wide runs were OS-pinned to one CPU; eight-wide runs used one hardware thread per physical core. Wall clock covers the whole solver process; core-seconds are measured user plus system CPU time, not wall time multiplied by requested width. Table 3 retains all three geode requests that recover the six reference modes. Two are tuned using the reference spectrum: requesting thirty modes at 4.5 GHz, or moving the shift to 20 GHz. The older port-aware extraction route does not use reference frequencies to choose its request, but returns two non-physical modes alongside the six; identifying the physical subset still uses Palace. Palace’s target is a lower bound, whereas geode’s shift is a centre, so identical numerical requests need not select identical spectra.

At eight wide Palace without field output takes 23.64s, less than any of these geode routes (27.26–51.12s). With field output, Palace takes 33.35s and the two oracle-tuned routes are faster. At one wide the geode routes take less wall and CPU time, but neither comparison isolates a solver kernel: geode writes no fields and has no counterpart to Palace’s error estimator, which remains enabled even with Save=0 (about 40.2s at one rank and 5.0s at eight). MPI busy-waiting also affects measured Palace core-seconds. No single route is selected as a headline winner.

The Palace build uses SuperLU-preconditioned GMRES, not AMS: the pinned source and build configuration resolve `Default` to SuperLU. A later, separate m6i session with a tuned Palace build found only a 1.01–1.05× same-box wall-time improvement over its untuned control; cross-session differences must not be credited to compiler tuning. Those runs also predate the new port-subspace solver. The local update in Section 6.2 cannot be substituted into this same-box timing table.

At 1,157,564 DOFs, a historical geode direct run completed on a 128 GB box in 565.5s with 92.2 GB peak RSS. Palace’s earlier eight-rank run recorded 423.12s and about 33 GB aggregate memory. These are different-host observations, and the geode request lacks mode-content verification. They establish the direct factorization’s large memory requirement, not a controlled performance crossover or equivalent physical-mode throughput.

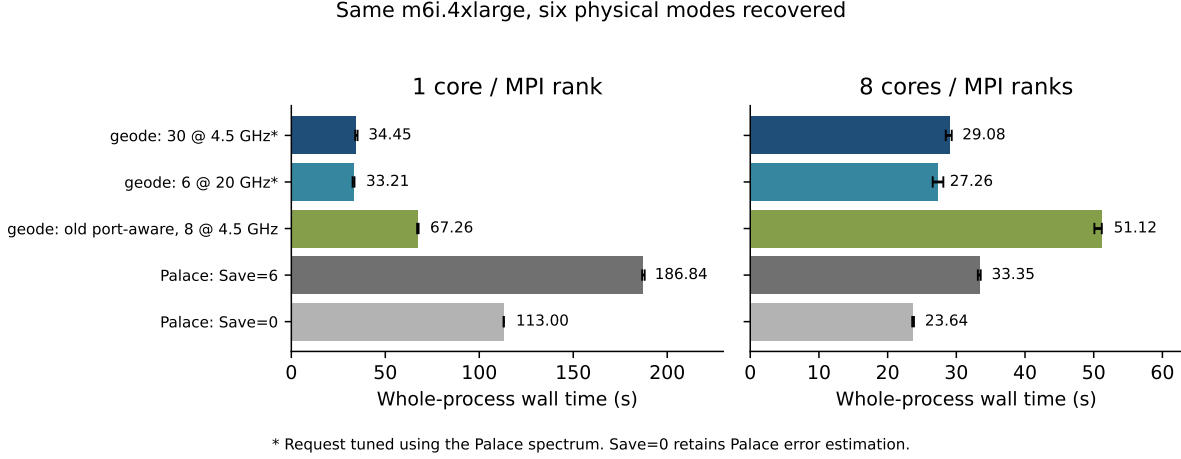


Figure 6: All three mode-recovering geode requests beside Palace with and without field output, from Table 3. Bars show medians; whiskers show the observed minimum and maximum. The oracle-tuned requests are not a reference-free recipe for an unknown device. Save=0 removes Palace’s field output, but retains its error estimator.

Table 4: Driven CPU-f64 linear-solve seconds on one r6i.4xlarge, one core/rank, median of two geode and three Palace runs, except 897,650 edges (one run each). geode includes operator assembly, preconditioner setup, iteration and residual check; Palace sums its Setup, Preconditioner and Linear Solve timers, excluding Operator Construction. These are not whole-process timings.

Edges	geode Jacobi	geode AMS	Palace AMS	Palace SuperLU
25,695	2.4	0.7	1.0	1.4
102,024	22.2	5.3	5.0	8.9
462,520	227.3	56.4	26.4	85.9
897,650	624.3	181.3	53.3	235.8

11.2 CPU driven solve: AMS at scale

A separate f64 experiment uses assembled COCG on the lossy driven cube at $\omega = 0.10$, on one dedicated **r6i.4xlarge**, with geode **64d44269**. AMS reduces one-core linear-solve time relative to Jacobi by 3.4–4.2 $\times$ across 25,695–897,650 edges. It reduces iterations from 2969–13662 to 39–105, but does not eliminate iteration growth. Every measured solve meets an explicitly recomputed relative-residual tolerance of 10^{-8} .

At 462,520 edges AMS needs about 3.4 GB versus Jacobi’s 1.05 GB. At 897,650 edges it needs 8.5 GB versus 2.0 GB, and its setup alone takes 53.488 s. Eight threads reduce setup but do not accelerate geode’s Krylov iteration; at 462,520 edges its AMS solve takes 50.4 s versus Palace AMS’s 4.2 s on eight ranks. The one-core geode AMS advantage at the smallest size does not persist at scale: Palace AMS takes 53.3 s against 181.3 s at the largest size. These measurements concern one structured first-order fixture, assembled CPU operators and low frequencies. They establish neither GPU performance nor a remedy for the interior eigenproblem.

11.3 GPU cell: correctness proven, scaling an honest negative

geode-fem’s GPU path accelerates the *driven* solve (matrix-free apply + on-device COCG), not the eigensolve of the credential, and all GPU execution is CUDA-f32 (Section 3). On 2026-07-14, geode-fem code executed on a physical GPU for the first time (EC2 **g6e.xlarge**, 1 $\times$ NVIDIA L40S 46 GB,

Table 5: Driven-solve wall-clock scaling, solve-only medians of $n = 3$ timed repetitions (warm-up excluded), same host (`g6e.xlarge`: 4 vCPU + 1× L40S). One fixture (σ -lossy lumped-port cube) at fixed ω , mesh-refined across the columns. f64 iterative configurations at tolerance 10^{-8} ; the f32 GPU configuration at 10^{-6} (its recurrence floor sits above 10^{-8}). The accuracy row is the full-field relative L_2 error of the GPU-f32 solution against the Direct-f64 reference. At 25,695 edges the GPU-f32 configuration converged in the committed single-frequency runs but not in its five-point frequency sweep, and run-to-run convergence is nondeterministic at that size (f32 floor). Values verbatim from the committed `benchmarks/gpu_driven_scaling/results.toml`.

Configuration	Mesh size (Nédélec edges)			
	1,854	5,859	13,428	25,695
Direct faer LU (CPU f64), s	0.024	0.203	1.540	6.036
Assembled-CSR COCG (CPU f64), s	0.032	0.198	0.709	1.865
Matrix-free COCG (CPU f64), s	1.652	8.885	30.234	82.056
Matrix-free COCG (GPU f32), s	4.388	13.351	34.381	81.764
GPU-f32 accuracy vs. Direct-f64 (rel. L_2)	1.26×10^{-4}	2.93×10^{-4}	4.60×10^{-4}	1.20×10^{-3}

driver 595.71.05, CUDA 13.2): the two CUDA-f32 correctness smokes — the matrix-free Nédélec matvec and the on-device COCG — pass against the f64 CPU reference within f32-appropriate tolerances (15.0 and 3.3 s wall, durations dominated by GPU init and kernel JIT — correctness evidence, not performance).

The scaling study (committed as `benchmarks/gpu_driven_scaling/results.toml`; a σ -lossy lumped-port cube fixture at fixed frequency, mesh-refined from 1,854 to 25,695 edges, all sixteen cells on the same host) is an honest negative: **GPU-f32 matrix-free loses to every CPU configuration at every measured size** (Table 5). Computed from the artifact’s unrounded medians, the assembled-CSR COCG on CPU is $136\times$ faster at 1,854 edges (displayed: 0.032 versus 4.39 s) and still $44\times$ faster at 25,695 (1.86 versus 81.8 s); even the direct LU is $13.5\times$ faster at the top size (6.04 versus 81.8 s). The mechanism is kernel-launch domination (~ 28 ms per iteration at the top size versus ~ 0.63 ms for assembled-CSR on CPU) — 25.7k edges is far too small to fill an L40S — and f32 accuracy compounds the verdict (relative error degrading to 1.2×10^{-3} with refinement; nondeterministic convergence at the top size). One directional positive survives, and it is the one the architecture predicts: against the *same matrix-free algorithm on CPU*, the GPU’s deficit falls monotonically to parity at ~ 26 k edges (81.8 versus 82.1 s). We state the GPU position as trajectory, not achievement, with tracked levers (kernel fusion; on-device f64/mixed-precision when cubecf ships it; preconditioning) — and we prefer publishing the bound to implying the win. This cell is a cube fixture family, not the transmon eigenproblem, and its CPU baselines (4-vCPU host) are not comparable to the `m6i` numbers of Table 3.

12 Reproducibility

Every artifact behind every number in this paper is committed and scripted.³

- **Geometry and mesh.** Generated by DeviceLayout.jl v1.15.0’s `SingleTransmon` example (AWS Center for Quantum Computing, 2025); the fixture generator is committed, and the MSH 4.1 fixture is pinned by SHA-256 (5b3ff4c3...b33dd; full hash and provenance in the committed `transmon_smoke.provenance.txt`).
- **The MSH conversion gotcha.** MFEM’s MSH 4.1 reader rejects the DeviceLayout multi-entity-block file, so the Palace runs consume an MSH 2.2 conversion (`gmsh -save -format`

³Repository URL and archival DOI for the geode-fem source to be added at submission. TODO(operator).

msh2). The conversion is node-preserving (all 22,684 nodes unchanged, physical groups preserved) and physics-neutral: eigenvalues reproduce bit-for-bit across the conversion. Documented in the committed `palace_config.provenance.txt` because it is exactly the kind of silent pipeline step a same-mesh claim must disclose.

- **Credential artifacts.** The Palace configuration and provenance under `reference/fixtures/transmon_palace/`; geode-fem’s benchmark definitions, the agreement numbers, the L-doubling tripwire (17.4901 $\rightarrow$ 12.37 GHz, committed ratio 0.7071), the spurious-mode measurements (3.4528 GHz, $p = 0.9942$; L-doubling ratio 0.7081), and the full gauge/projection arc of Section 6.2 (tree-cotree counts and the 1.64% shift; bulk-projection divergence ratio 50.2 and projected norm 1.06×10^{-4} ; port-aware acceptance at $\leq 0.029\%$) in `benchmarks/transmon_eigen/results.toml`, each pinned by a release-tier regression test. Palace’s raw outputs (`eig.csv`, per-port participation, run log) are committed; a rerun reproduced `eig.csv` bit-for-bit.
- **Precision of the committed agreement numbers.** `benchmarks/transmon_eigen/results.toml` stores geode-fem frequencies at three decimals while the Palace column carries full precision, so the committed per-mode `rel_err_pct` is computed against the rounded geode-fem values; at full precision the agreement is slightly better (approximately 0.029% worst case). We quote the committed-artifact numbers throughout and disclose the rounding here. Derived ratios quoted in the text (core-second and speedup multipliers) are computed from the displayed table values unless marked as committed.
- **LOM chain.** The three-conductor capacitance extraction and Koch-exact spectrum ($C_\Sigma = 136.7$ fF, $E_C = 0.142$ GHz, and the anchor-gap diagnosis of Section 7) in `benchmarks/transmon_quantum/results.toml`; pinned by the release test `crates/geode-core/tests/transmon_quantum.rs`.
- **Adjoint layer.** The four sensitivity modules with their FD-and-mutation test suites: `crates/geode-core/src/adjoint.rs` (PR #573); `crates/geode-core/src/shape.rs` (PRs #575/#586; this module also carries `capacitance_shape_gradient`); `crates/geode-core/src/driven/adjoint.rs` (PR #579); and `crates/geode-core/src/driven/shape.rs` (PR #581).
- **Optimization demonstrations.** The parallel-plate loop proof (full trajectories, closed-form check, fresh-solve confirmation) in `benchmarks/transmon_diffopt/results.toml`; the real-device FD validation, demonstration convergence, and the bounded anchor attempt with its mesh-safety budget in `benchmarks/transmon_diffopt/pad_results.toml`; harmonic motion and its bounded anchor attempt in `benchmarks/transmon_diffopt/harmonic_results.toml`; all consumed by `crates/geode-core/tests/transmon_diffopt.rs`.
- **Performance artifacts.** The historical CPU cell in `benchmarks/transmon_bench_cpu/results.toml`; mode-verified timing in `benchmarks/transmon_bench_cpu/results_like_for_like_m6i.toml`; tuned-build controls in `benchmarks/transmon_bench_cpu/results_tuned_palace_m6i.toml`; local port-subspace evidence in `benchmarks/transmon_bench_cpu/results_like_for_like_local.toml`; driven CPU AMS results in `benchmarks/gpu_driven_scaling/results_ams_cpu_r6i.toml`; the 1.16M-DOF completion run log at `benchmarks/transmon_bench_cpu/geode_runs_1p16M_2026-07-15.log`; the GPU scaling cell (all sixteen cells, iteration counts, recomputed true residuals, DNF handling, same-host provenance) in

benchmarks/gpu_driven_scaling/results.toml; the deep-shift eigensolve characterization at benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md.

- **Hardware disclosure.** CPU cell: EC2 `m6i.4xlarge` (8 physical cores / 16 vCPU, 64 GB), `us-east-1` for the mode-verified re-timing (the historical session was in `us-west-2`); driven AMS: `r6i.4xlarge`, `us-east-1`; the 1.16M-DOF completion row: a 128 GB `r6i`-class box (see the committed log). GPU cell: EC2 `g6e.xlarge` (4 vCPU, 30 GB; 1× NVIDIA L40S 46 GB; driver 595.71.05, CUDA 13.2).
- **Versions.** geode-fem commit 3174015 (headline credential and historical CPU cells; the adjoint/diffopt layers are merged post-3174015 and cited by PR number); Palace commit `fba6a5b`; DeviceLayout.jl v1.15.0; Burn/burn-cuda 0.21.

13 Discussion

What the credential does and does not establish. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) the cross-validation is code verification and code-to-code comparison, not validation against nature: two independent implementations agreeing to $\leq 0.033\%$ on the identical discrete problem, with analytic tripwires pinning the junction physics. It does not certify that either solver matches a measured device — that is what Ye et al. (2025)-style experimental benchmarks address — but it rules out the large error class living in formulation, assembly, boundary handling, and eigensolver machinery. For this paper’s purpose it establishes the specific premise the gradients need: the forward solutions being differentiated are production-accurate.

What the differentiable-design evidence supports. The proven capability is bounded and we state its boundary. Proven: the 2×2 adjoint matrix at operator level (FD-validated with mutation tests); the capacitance $\rightarrow E_C$ chain with its stationarity collapse; a closed Newton loop confirmed by independent fresh solves on both an analytic fixture and the real 133k-tet mesh; and a real-mesh gradient validated at 1.15×10^{-4} against the full independent pipeline. Not proven, and not claimed: eigenmode/EPR derivatives (the Section 10 wall); anchor-reaching real-device optimization (harmonic motion still stops at 128.793 fF in Section 9.2, with the multi-parameter result pending); tensor- ϵ shape chains; multi-parameter optimization at scale; and any wall-clock superiority over sweep-based workflows. The capability is also not qubit-specific — shape and material sensitivities of electrostatic and driven-EM observables apply to RF and photonic device design generally; the transmon is the demonstration vehicle with the clearest documented unmet need.

Threats to validity. A common-mode error — both solvers wrong the same way — cannot be excluded by cross-comparison alone; the analytic anchors and disjoint implementations make it unlikely for the junction physics. The optimization demonstrations are single-parameter by design; nothing here demonstrates multi-parameter or constrained optimization behavior. The parallel-plate tier’s one-step convergence is an affine-map property, disclosed as such. The real-device chain uses the scalar trace-averaged sapphire (tensor- ϵ delta quantified at 0.75%). Repeat counts and timing boundaries differ across experiments and are stated in each table. The large-scale eigenmode completion used a different 128 GB box and lacks mode-content verification; it cannot establish equivalent-work throughput.

Limitations. (i) One geometry, one mesh, first order only — higher-order and mesh-refinement convergence is future work (a $p = 2$ Palace configuration is committed for that purpose). (ii) The

lossless model leaves readout ports open; resistive terminations (hence complex-symmetric pencils and mode Q s) are out of scope. (iii) The credential eigenpath remains the ungauged committed benchmark; the new port-subspace route retains all six modes when seven are requested, but the solenoidal port artifact is still identified using the reference spectrum rather than eliminated. (iv) The CPU evidence depends on request selection, field output, error estimation and implementation version; it supports no unconditional per-core or scaling advantage. (v) GPU evidence covers driven-solve correctness and an honest scaling negative in f32 on a cube fixture family; no GPU eigensolve path exists. (vi) The differentiable branch is electrostatic/LOM only (eigenmode/EPR is roadmap), currently scalar- ϵ for the shape chain, and its real-device demonstration is bounded by the fixed-topology parametrization budget.

14 Conclusion

The discrete-adjoint layer supplies solver-derived material and geometry gradients for electrostatic and driven FEM operators. Its stationary capacitance derivative closes the geometry-to-charging-energy loop, with fresh-solve confirmation on an analytic fixture and a within-budget real-device target. Harmonic motion extends the real-mesh safe deformation budget $8.90\times$, but the 89.9 fF anchor remains unreached; the multi-parameter experiment is pending. Same-mesh agreement with Palace supplies a forward-solver credential, while the updated mode-verified timing and driven AMS experiments delimit the computational cost without claiming a general performance advantage. Eigenmode/EPR derivatives remain future work.

Acknowledgments. geode-fem’s development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline, with every reported number traced to a committed, human-auditable artifact.

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